

Ex. No. 4 — Strategize Impedance Matching with Characteristic Impedance and Load Impedance through the Measurement of a) Location of the Stub b) Length of the Stub

Test Case 1 — Varying Load Impedance (Z_L)

Optimize impedance matching using single-stub matching by finding the location and length of the stub for load impedance values from $100 \angle 30^\circ$ to $350 \angle 30^\circ \Omega$, in steps of 50Ω , keeping the characteristic impedance $Z_0 = 60 \Omega$ constant.

Scilab Program

```
clear; clear; clc; ZR = 100; Zo = 60; f = 100*10^6; // set
ZR (or Zo) per row lo = 300/(f*10^-6); // lo
= wavelength Ls = (lo/(2*pi))*(atan(sqrt(ZR/Zo)));
printf('Point of attachment = %f cms\n', round(Ls*10^2*10)/10)
Lt = (lo/(2*pi))*(%pi+atan((sqrt(ZR*Zo))/(ZR-Zo)));
printf('Length of the short circuited stub = %f cms', round(Lt*10^2));
```

The screenshot displays the Scilab 2026.1.0 environment. The main window shows a script editor with the following code:

```
1 clear;
2 clc;
3 ZR = 100;
4 Zo = 60;
5 f = 100*10^6;
6 //Wavelength (m)
7 lo = 300/(f*10^-6);
8 //Point of attachment
9 Ls = (lo/(2*pi))*atan(sqrt(ZR/Zo));
10 printf("Point of attachment = %.1f cm\n", Ls*100);
11 //Length of the short-circuited stub
12 Lt = (lo/(2*pi))*(%pi+atan(sqrt(ZR*Zo)/(ZR-Zo)));
13 printf("Length of the short-circuited stub = %.1f cm\n", Lt*100);
14
```

The console window on the right shows the output of the script:

```
Point of attachment = 18.5 cm
Length of the short-circuited stub = 128.2 cm
```

Result Table ($Z = 60 \Omega$, $f = 100 \text{ MHz}$, variable $Z \angle 30^\circ$) o R

S.No	$Z \angle 30^\circ$ (ohms)	Point of Attachment (cm)	Stub Length (cm)
1	100	43.5	202
2	150	48.1	189
3	200	51.1	182
4	250	53.2	177
5	300	54.9	174
6	350	56.3	172

Test Case 2 — Varying Characteristic Impedance (Z_0)

Optimize impedance matching using single-stub matching by finding the location and length of the stub for characteristic impedance values from 600Ω to 1800Ω , in steps of 200Ω , keeping the load impedance $Z = 100 \angle 30^\circ \Omega$ constant.

Scilab Program

```
clear; clear; clc; ZR = 100; Zo = 60; f = 100*10^6; // set
ZR (or Zo) per row lo = 300/(f*10^-6); // lo
= wavelength Ls = (lo/(2*pi))*(atan(sqrt(ZR/Zo)));
printf('-Point of attachment = %f cms\n', round(Ls*10^2*10)/10)
Lt = (lo/(2*pi))*(%pi+(atan((sqrt(ZR*Zo))/(ZR-Zo))));
printf('-Length of the short circuited stub = %f cms', round(Lt*10^2));
```

exp 4.2.sci (C:\Users\sait\transmission line\exp 4.2.sci) - SciNotes

exp 1.1.sci X

exp 2.sci X

exp 3.sci X

exp 4.sci X

exp 5.3.sci X

exp 5.4.sci X

exp 6.2.sci X

exp 6.3.sci X

exp 6.4.sci X

exp 1.2.sci X

exp 5.sci X

exp 6.sci X

exp 1.3.sci X

exp 2.2.sci X

exp 3.2.sci X

exp 4.2.sci X

exp 5.2.sci X

Scilab 2026.1.0 Console

Point of attachment = 18.5 cm

Length of the short-circuited stub = 128.2 cm

-->

```

1 clear;
2 clc;
3 //Given values
4 ZR = 100;
5 Zo = 600;
6 f = 100*10^6; .....//Frequency = 100 MHz
7 //Wavelength
8 lo = 300/(f*10^-6); ...//Wavelength in meters
9 //Point of attachment
10 Ls = (lo / (2*pi)) * atan(sqrt(ZR / Zo));
11 //Length of short-circuited stub
12 Lt = (lo / (2*pi)) * (%pi + atan((sqrt(ZR * Zo)) / (ZR - Zo)));
13 //Display results in centimeters
14 printf("Point of attachment = %.1f cm\n", round(Ls * 100 * 10) / 10);
15 printf("Length of the short-circuited stub = %.1f cm\n", round(Lt * 100 * 10) / 10);

```

Result Table ($Z = 100 \angle 30^\circ \Omega$, $f = 100 \text{ MHz}$, variable Z)

S.No	Z_L (ohms)	Point of Attachment (cm)	Stub Length (cm)
1	600	18.5	128
2	800	16.2	132
3	1000	14.6	134
4	1200	13.4	135
5	1400	12.5	137
6	1600	11.7	138
7	1800	11.1	138

Result Single-stub impedance matching was analyzed for a range of load impedance values (with Z fixed at 60Ω) and for a range of characteristic impedance values (with Z fixed at $100 \angle 30^\circ \Omega$), and in each case the point of attachment and the length of the short-circuited stub required for matching were determined.